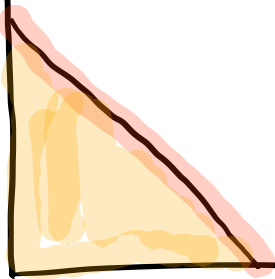


Review for Prelim 1

Budget Line

Affordable

Good
Y

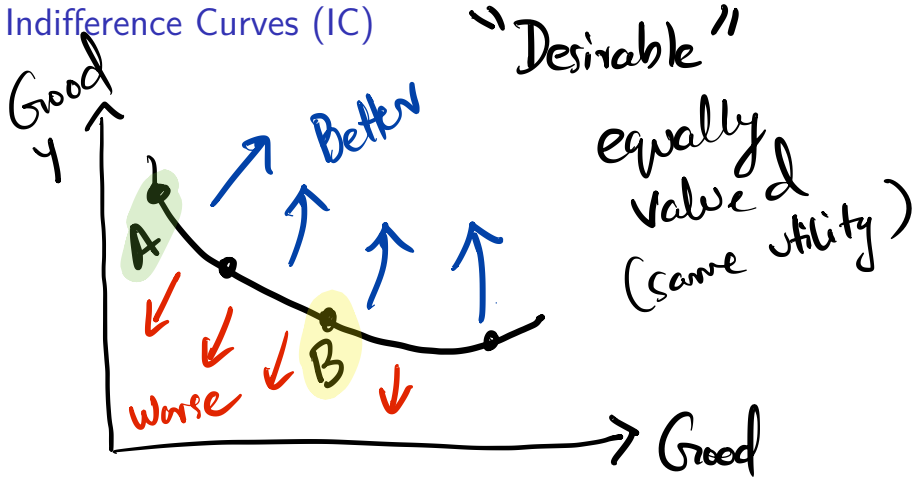


$$P_x X + P_y Y = \underline{M}_{\text{income}}$$

$$\text{slope} = -\frac{P_x}{P_y}$$

Good
X

Indifference Curves (IC)



$MRS_A \neq MRS_B$ indifferent $\times$

Marginal Utility & MRS



slope of the I.C

$$= -\frac{MU_x}{MU_y}$$

↓
change in utility
when one good
is changed by
one unit

$$MU_x = \frac{\Delta U}{\Delta X}$$

Y constant

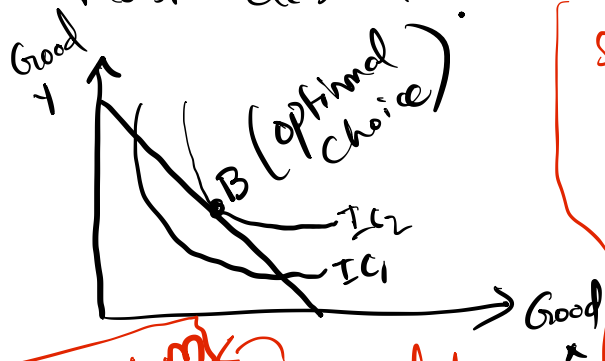
$$MU_y = \frac{\Delta U}{\Delta Y}$$

X constant

Consumer Optimum

B.L. $\rightarrow$ afford
I.C. $\rightarrow$ "desire" (highest I.C.)

Among the affordable bundles, which is the most desired?



slope of I.C. = slope of B.L.

$$MRS = -\frac{P_x}{P_y}$$

$$-\frac{MU_x}{MU_y} = -\frac{P_x}{P_y}$$

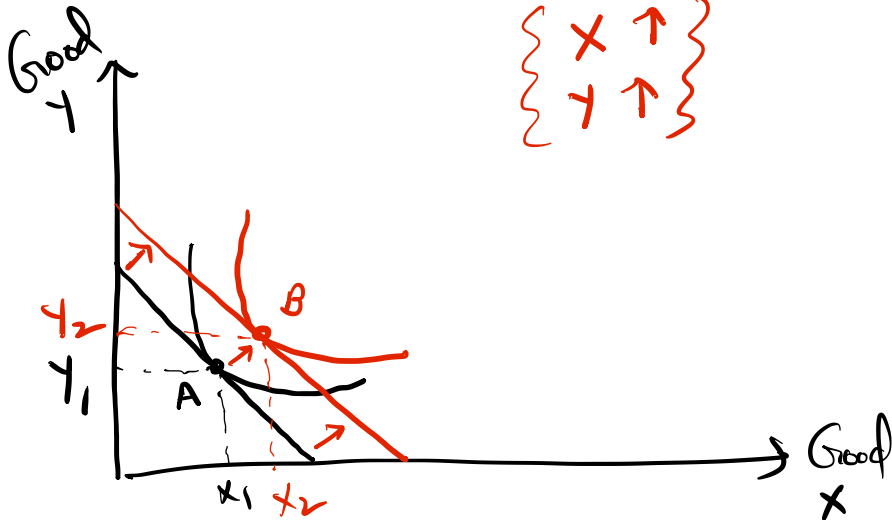
$P_x X + P_y Y = m$ $\rightarrow$ Optimality condition $\rightarrow \frac{MU_x}{MU_y} = \frac{P_x}{P_y}$

Consumer Optimum

Income changes

Income ↑ Yes.

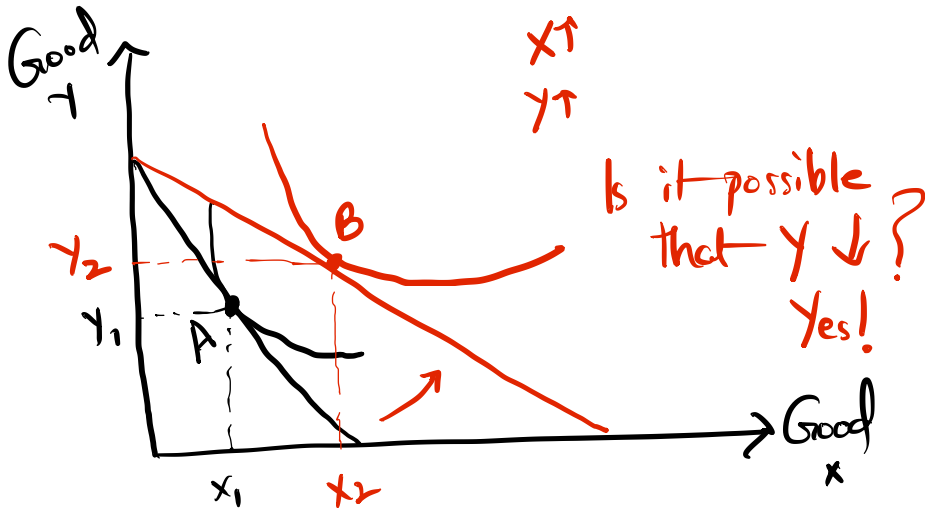
$\left\{ \begin{matrix} X \uparrow \\ Y \uparrow \end{matrix} \right\}$



Consumer Optimum

Change in price of a good

$P_x \downarrow$ ses. $\{ \text{PIVOT} \}$



Price changes

Decomposing into substitution and income effects

$P_x \downarrow$

$X \uparrow$

$Y \downarrow$

"richer"

$X \uparrow$

$Y \uparrow$

Total effect = S.E. + I.E.

= $X \uparrow$

$Y (?)$

Stronger

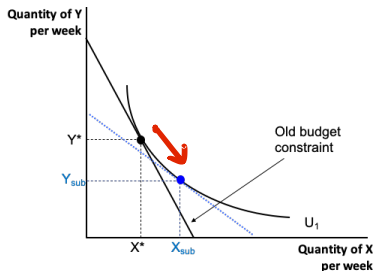
Price changes

Decomposing into substitution and income effects

Substitution effect

New prices, Old Utility

$P_x \downarrow$, which means x has become relatively cheaper than y , making the consumer substitute away from y towards x .



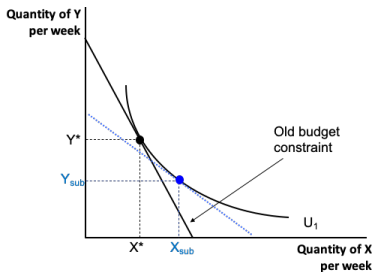
Price changes

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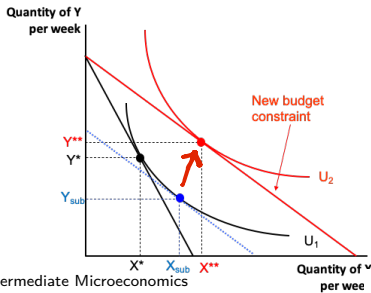
$P_x \downarrow$, which means x has become relatively cheaper than y , making the consumer substitute away from y towards x .



Income effect

New prices, New Utility

$P_x \downarrow$, which means that the individual now feels like they have extra income (*richer*), which means that they buy more of both x and y .



Price elasticity of demand $= \frac{\% \Delta Q}{\% \Delta P} .$

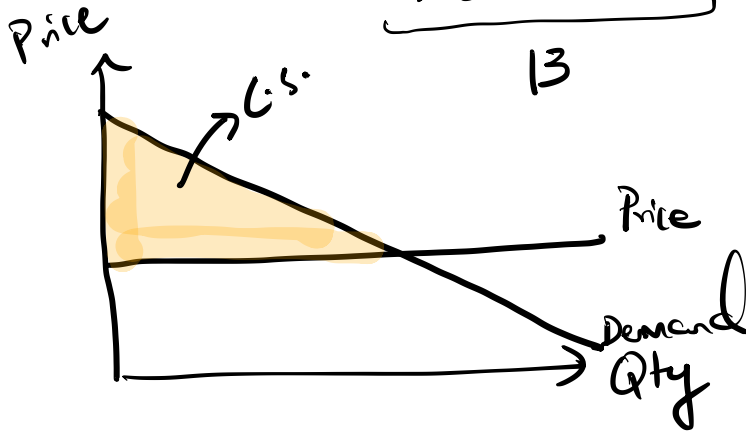
-2, -5 $\rightarrow$ relatively elastic

-0.2, -0.4 $\rightarrow$ relatively inelastic.

Price elasticity of demand

Consumer Surplus

$$= \underbrace{\text{WTP}}_{25} - \underbrace{\text{Price paid}}_{12} = 13$$



Production

We now shift our attention to decisions related to firm behavior (production).

Production function

Some basics

A production function is a mathematical relationship between inputs used and outputs produced.

Quantity produced is a function of the inputs used.

$$Q = f(K, L, M, N, \dots)$$

Production function

Some basics

A production function is a mathematical relationship between inputs used and outputs produced.

Quantity produced is a function of the inputs used.

$$Q = f(K, L, M, N, \dots)$$

Like in consumer theory we included two goods in the utility function, we consider two inputs in the production function – labor (L) and capital (K).

Total Product

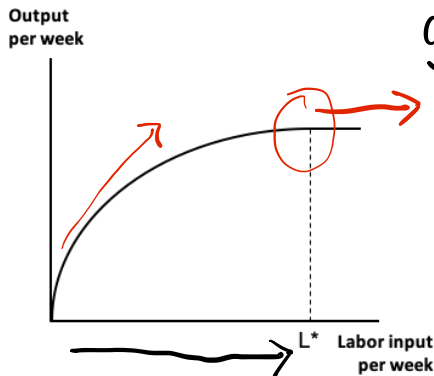
Akin to utility function (production function)

Total product is the amount of output produced when a certain combination of different inputs is used.

Total Product

Akin to utility function (production function)

Total product is the amount of output produced when a certain combination of different inputs is used.



Marginal Product

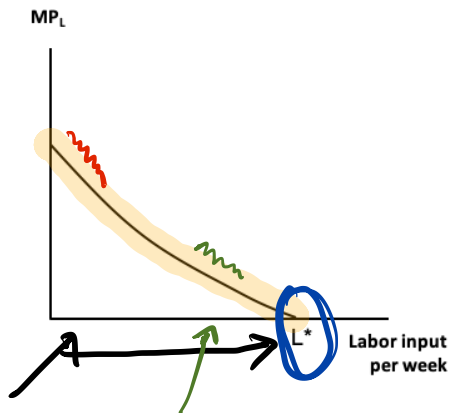
↪ $\frac{\Delta TP}{\Delta \text{input}}$

all other inputs constant.

Marginal Product

Marginal product of an input is the additional output produced when one more unit of the input is used.

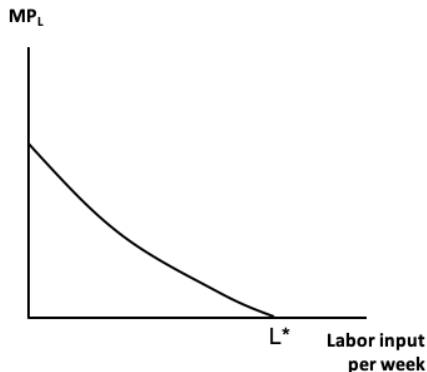
Example: Marginal product of labor is equal to the change in output when we add one more unit of an input (labor), while keeping the amount of the other input (capital) constant.



Marginal Product

Marginal product of an input is the additional output produced when one more unit of the input is used.

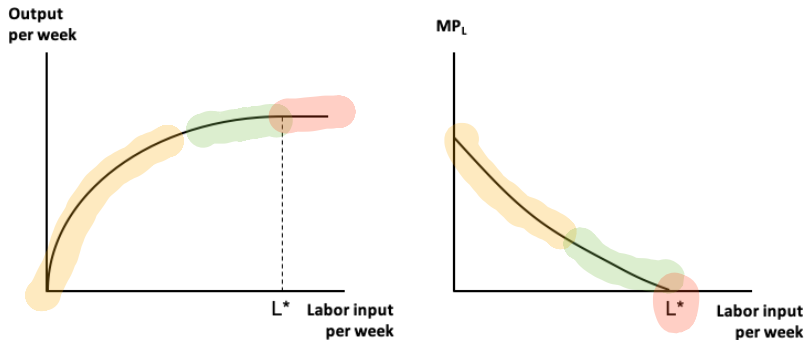
Example: Marginal product of labor is equal to the change in output when we add one more unit of an input (labor), while keeping the amount of the other input (capital) constant.



Akin to marginal utility.

Marginal Product & Total Product

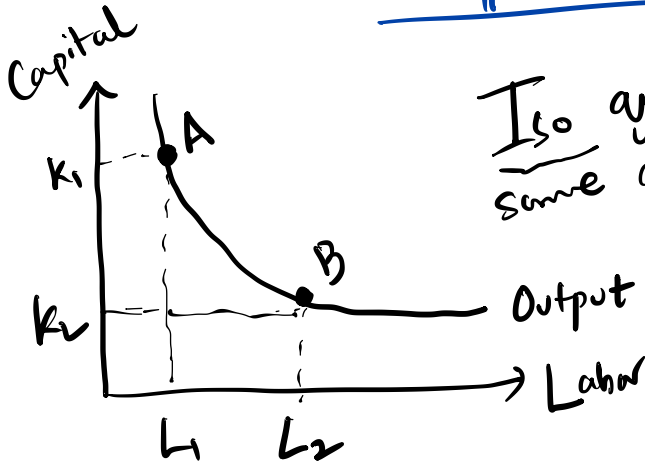
Marginal Product & Total Product



Isoquant

Iso means "same".

Indifference Curve



Iso quant
same quantity

Isoquant

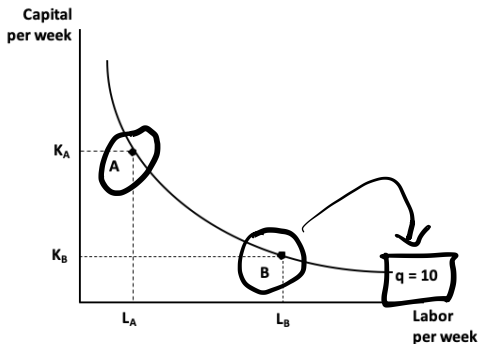
Iso means "same".

Isoquant is a curve that shows the various combinations of inputs that will produce the same amount of output.

Isoquant

Iso means "same".

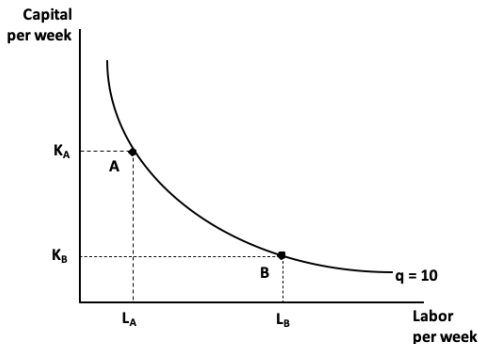
Isoquant is a curve that shows the various combinations of inputs that will produce the same amount of output.



Isoquant

Iso means "same".

Isoquant is a curve that shows the various combinations of inputs that will produce the same amount of output.



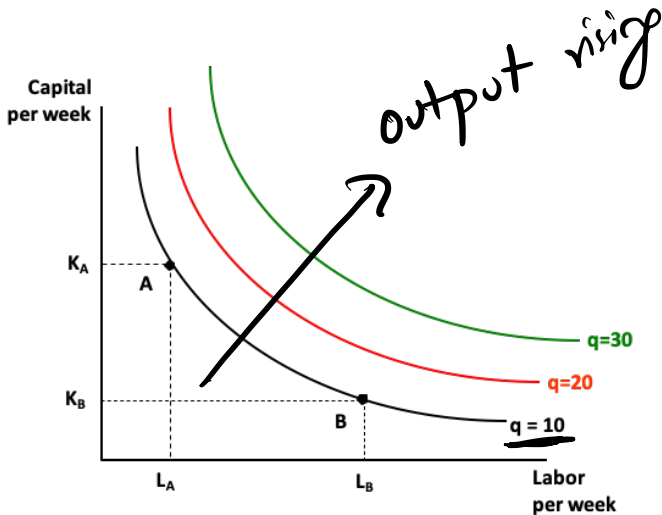
Isoquant is akin to indifference curve. Like the indifference curve, isoquants are also negatively sloped. (Think about why that should be the case!)

Isoquant

Increasing output levels.

Isoquant

Increasing output levels.



Marginal rate of technical substitution (MRTS)

Akin to Marginal Rate of Substitution (MRS)

$$MRS = \text{slope of I.C.} = -\frac{MU_x}{MU_y}$$

$$MRTS = \text{slope of I.Q.} = -\frac{MP_x}{MP_y} = -\frac{MP_L}{MP_K}$$

L, K

Marginal rate of technical substitution (MRTS)

Akin to Marginal Rate of Substitution (MRS)

MRTS is the amount by which capital (K) can be reduced when one more unit of labor (L) is added while holding output constant.

Marginal rate of technical substitution (MRTS)

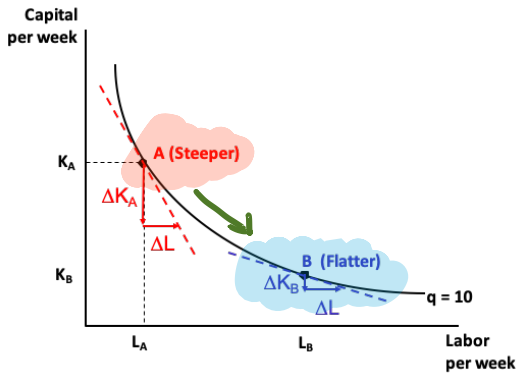
Akin to Marginal Rate of Substitution (MRS)

MRTS is the amount by which capital (K) can be reduced when one more unit of labor (L) is added while holding output constant. It is equal to the negative of the slope of the isoquant curve.

Diminishing MRTS

Akin to diminishing MRS

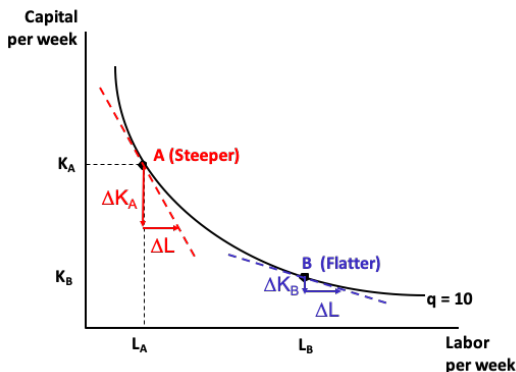
$$MRTS_A > MRTS_B$$



Diminishing MRTS

Akin to diminishing MRS

$$MRTS_A > MRTS_B$$

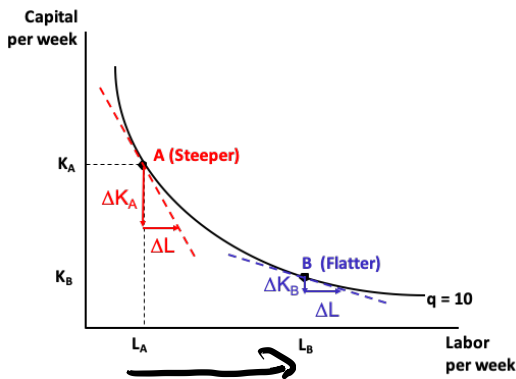


Slope of isoquant falls (like that of the indifference curve) – why?

Diminishing MRTS

Akin to diminishing MRS

$$MRTS_A > MRTS_B$$



Slope of isoquant falls (like that of the indifference curve) – why?

As the firm uses more and more labor, the $MP_L \downarrow$. Alternatively, the less (more) labor is used relative to capital, the more (less) able labor is able to replace capital in production.

What happens to isoquant when technology improves?

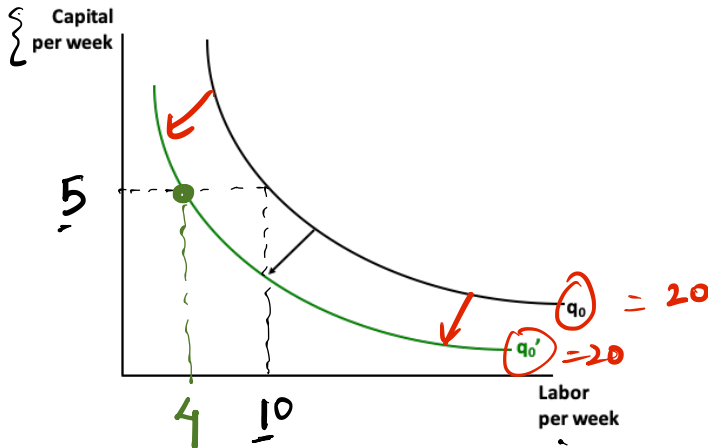
same inputs
more output

less inputs
same/more output.

What happens to isoquant when technology improves?

The technology improves $\Rightarrow$ the isoquant shifts from $q_0 \rightarrow q'_0$.

Note, both q_0 and q'_0 lead to the same level of output.



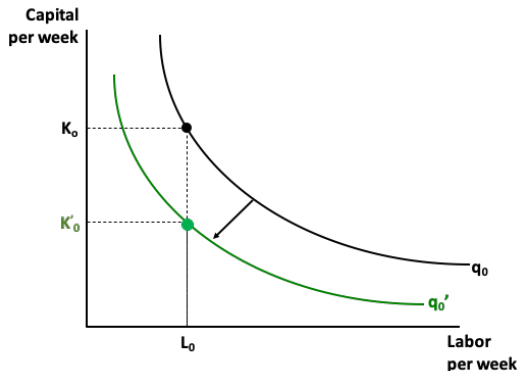
What happens to isoquant when technology improves?

Isoquant shifts to the left

The technology improves $\implies$ the isoquant shifts from $q_0 \rightarrow q'_0$.

Note, both q_0 and q'_0 lead to same level of outputs.

But q'_0 uses less capital (with same amount of labor) as compared to q_0 .



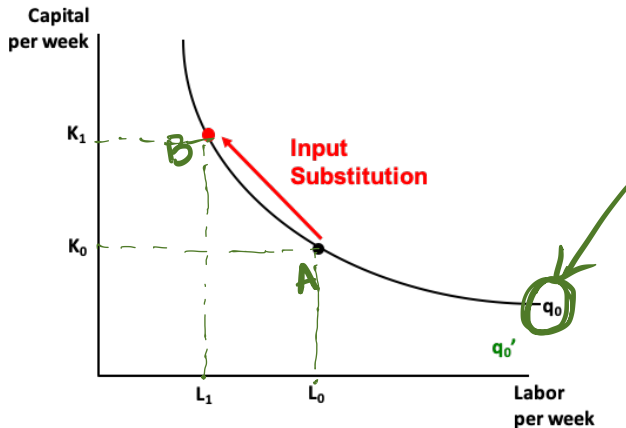
Input Substitution vs. Change in technology

↓
move along
the I.Q.

Shift of I.Q.

Input Substitution vs. Change in technology

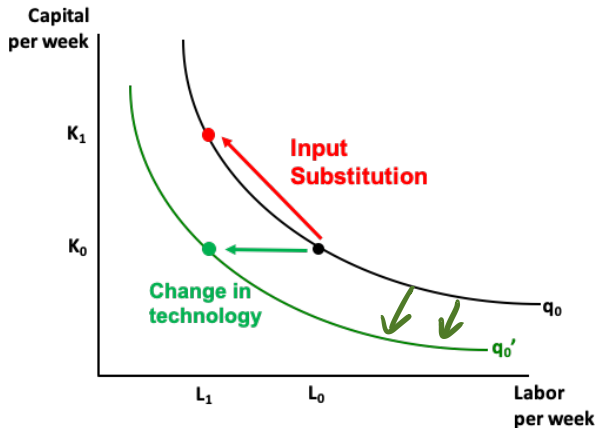
Input Substitution implies movement along same isoquant.



Input Substitution vs. Change in technology

Input Substitution implies movement along same isoquant.

Change in technology implies a shift in the isoquant.



Example

Question: The production function for frisbees is given by $q = 2K + L$, where q is the quantity of frisbees produced, K is the capital input and L is the labor input.

- If $K = 10$, then how much L is needed to produce 100 frisbees?
- If $K = 25$, then how much L is needed to produce 100 frisbees?
- Graph the isoquant for output of 100 frisbees, and indicate the points in parts a and b on the graph.
- Graph the isoquants for $q = 50$ and $q = 200$ for this production function.

Solutions

Question: The production function for frisbees is given by $q = 2K + L$, where q is the quantity of frisbees produced, K is the capital input and L is the labor input.

Part a

a. If $K = 10$, then how much L is needed to produce 100 frisbees?
Combining $K = 10$ and $q = 100$ with the fact that production function is given by $q = 2K + L$, we get then $100 = 2(10) + L$.
Therefore, $L = 100 - 20 = 80$.

Part b

b. If $K = 25$, then how much L is needed to produce 100 frisbees?
Combining $K = 25$ and $q = 100$ with the fact that production function is given by $q = 2K + L$, we get then $100 = 2(25) + L$.
Therefore, $L = 100 - 50 = 50$.

Solutions

c. Graph the isoquant for output of 100 frisbees, and indicate the points in parts a and b on the graph.

The equation for the given isoquant is $100 = 2K + L$. Therefore, it is clear that this is a linear isoquant, implying that it will be a straight line.

An easy way of plotting a linear isoquant line is to find the x-intercept (where the line meets the x-axis), and the y-intercept (where the line meets the y-axis).

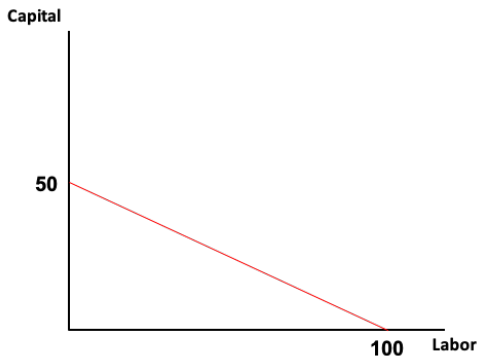
x-intercept is given by substituting $y = 0$. Since K is measured on the y-axis, we put in $K = 0$ in $100 = 2K + L$. We get that $L = 100$. Therefore, x-intercept is 100.

y-intercept is given by substituting $x = 0$. Since L is measured on the y-axis, we put in $L = 0$ in $100 = 2K + L$. We get that $K = 50$. Therefore, y-intercept is 50.

Solutions

c. Graph the isoquant for output of 100 frisbees, and indicate the points in parts a and b on the graph.

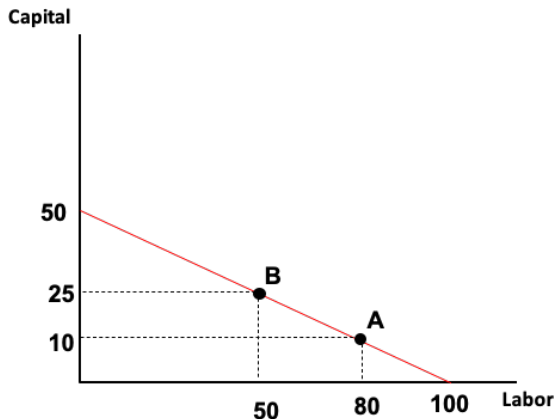
We have x-intercept is 100, and y-intercept is 50. Therefore, the isoquant takes the following form



Solutions

c. Graph the isoquant for output of 100 frisbees, and indicate the points in parts a and b on the graph.

And adding the points that we obtained from parts A and B.



Solutions

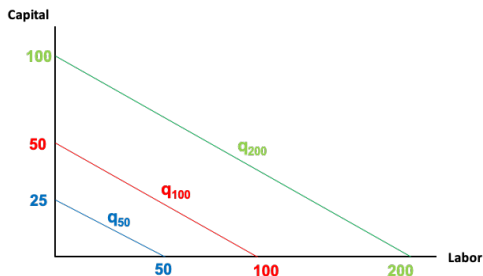
d. Graph the isoquants for $q = 50$ and $q = 200$ for this production function.

Part d

To plot the isoquants for $q = 50$ and $q = 200$, we follow the same process as above.

For $q = 50$, the x-intercept is 50, and the y-intercept is 25.

For $q = 200$, the x-intercept is 200, and the y-intercept is 100.



Solutions

e. Suppose, there is technological progress and the new production function for frisbees is given by $q = 3K + 1.5L$. Now answer parts a to d based on this new production function.

Try Yourself!

Returns to scale

Returns to scale

It is the rate at which output changes in response to a proportional change in all inputs.

Helps address the following question: Isn't it always better to operate/produce at a larger scale?

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Not necessarily. There are two opposing forces at work:

Returns to scale

It is the rate at which output changes in response to a proportional change in **all** inputs.

Helps address the following question: Isn't it always better to operate/produce at a larger scale?

Not necessarily. There are two opposing forces at work:

- ▶ Greater division of labor (specialization).
- ▶ Managerial inefficiencies and coordination problems

Constant returns to scale

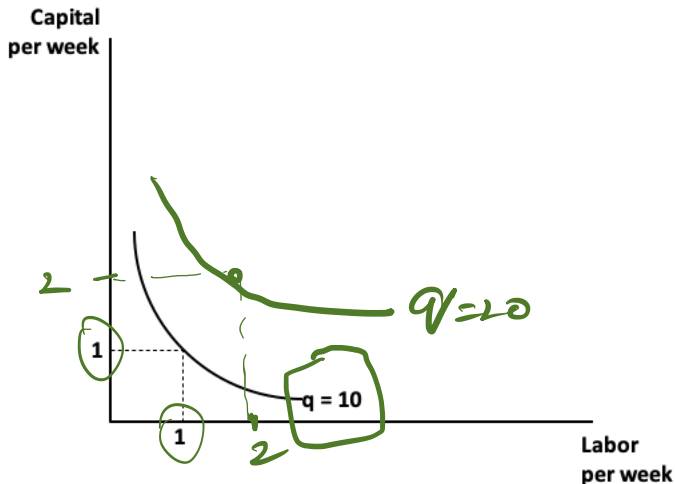
Output increases at the same proportion as inputs

inputs
↑ 20%.

output
↑ 20%.

Constant returns to scale

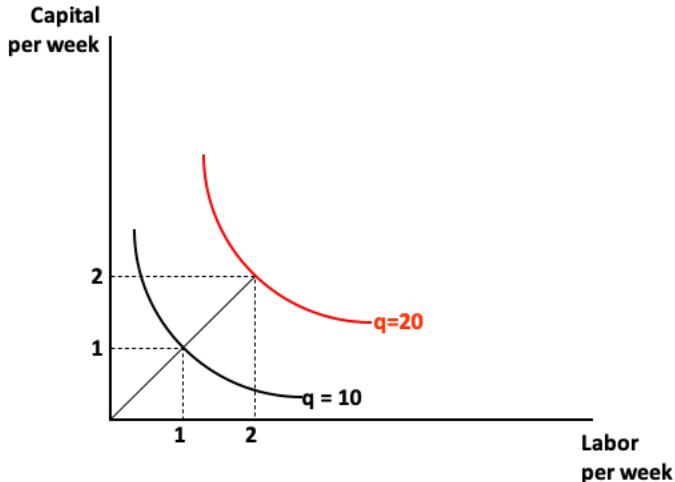
Output increases at the same proportion as inputs



Constant returns to scale

Output increases at the same proportion as inputs

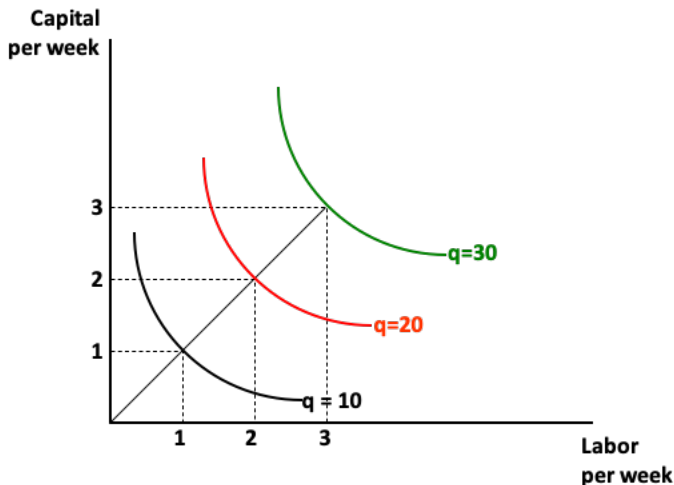
When inputs are doubled, then output is doubled.



Constant returns to scale

Output increases at the same proportion as inputs

When inputs are tripled, then output is tripled.



Increasing returns to scale

Output increases at a greater rate than inputs

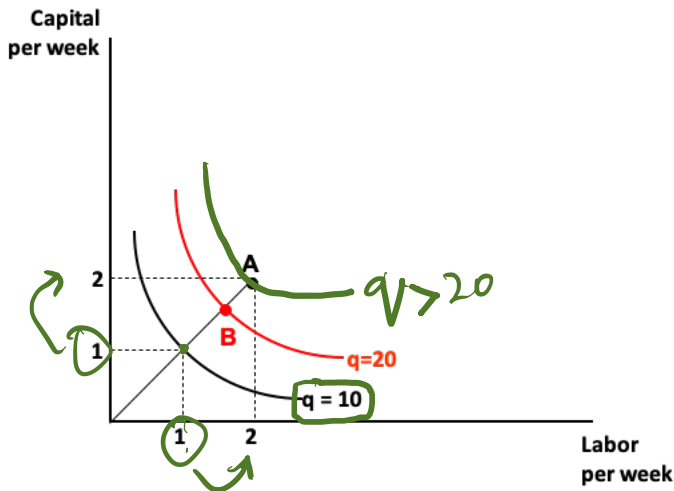
inputs
20%

output
> 20%

Increasing returns to scale

Output increases at a greater rate than inputs

Output is doubled when inputs are less than doubled.



Decreasing returns to scale

Output increases at a lower rate than inputs

Output is doubled when inputs are more than doubled.

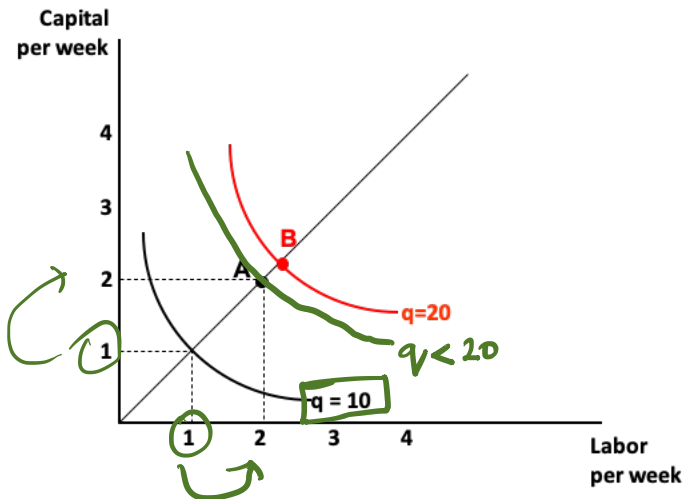
inputs
 $\uparrow 20\%$

output
 $< \uparrow 20\%$

Decreasing returns to scale

Output increases at a lower rate than inputs

Output is doubled when inputs are more than doubled.



Example

Returns to scale

Question: Suppose the production function is given by $q = AK^aL^b$, where A , a and b are all greater than zero, and q is the quantity produced, K is the capital input and L is the labor input.

This type of a production is called a **Cobb-Douglas** production function

If $a + b = 1$, then what can you infer about returns to scale?

I.R.S. D.R.S. C.R.S.

Solution

$$q = A L^a K^b$$

$$Land\ k = 2x.$$

$$\begin{aligned} q &= A \cdot (2L)^a \cdot (2K)^b \\ &= A \cdot 2^a L^a \cdot 2^b K^b \\ &= \underbrace{2^a \cdot 2^b}_{a+b} \cdot \underbrace{A L^a K^b} \end{aligned}$$

$$\rightarrow = 2^{a+b} q$$

$$q' = 2^{a+b} \cdot q$$

$$q' = 2 \cdot \underbrace{q}_{\text{old output}}$$

twice

C.R.S.

Solution

Solution

To make inferences about returns to scale, we need to find out what happens to output if all inputs are raised in the same proportion.

Lets assume that all inputs are doubled. Then we examine what happens to output.

$$\begin{aligned}q' &= A (2K)^a (2L)^b \\q' &= A \ 2^a K^a \ 2^b L^b \\q' &= 2^{a+b} \underbrace{A K^a L^b}_q \\q' &= 2^{a+b} q \\q' &= 2q\end{aligned}$$

Since when all the inputs are doubled then output also doubles in quantity, we can conclude that there are **constant returns to scale**.

Example 2

Try this yourself!

Question: With the same cobb-douglas production function ($q = AK^aL^b$) as before, what can be inferred about returns to scale in the following cases:

(a) $a + b > 1$

(b) $a + b < 1$

Hint: Follow the same process as described in the previous example, and then make a different substitution in the last step.

Example 3

Problem 5 from the book.

"Virtually every production function exhibits diminishing returns to scale because all inputs have diminishing marginal productivity. So when all inputs are doubled, output must less than double." True or False? Explain.

Example 3

Solution

Example 3

Solution